1 Students Exams Score Analysis

You are preparing a data analysis report for a classroom dataset. The goal is to analyze student performance using vectors and basic statistics, then present it professionally in LaTeX. This aligns directly with data science practice: organizing data, computing summaries, and presenting results clearly.

Dataset:

Student	Math	Phsic	Chem
Alice	90	85	92
Bob	78	82	88
Charlie	85	89	84
Diana	92	95	90
Ethan	70	75	80

Table 1: Dataset table

This is the computed and converted table:

Students	Marks Vector	Total T_i	Average m_i
Alice	$M_{Alice}\,[90, 85, 92]^T$	267	89.0
Bob	$M_{Bob}\,[78, 82, 88]^T$	248	82.67
Charlie	$M_{Chalie}\,[85, 89, 84]^T$	258	86
Diana	$M_{Diana}\,[92, 95, 90]^T$	277	92.33
Ethan	$M_{Ethan}\,[70, 7580]^T$	225	75

Finally, standard deviation for each student is computed by the following formula:

$$\sigma_i = \sqrt{\frac{1}{3}\sum_{j=1}^{3}(m_{ij} - m_i)^2}$$